24UAM511 · Natural Language Processing

Question Bank — Second half of Unit III, Unit IV and Unit V

AI & ML Engineering · III Year · Part A: 20 × 2 Marks · Part B: 12 × 16 Marks

PART A — 2 Mark Questions

Unit III (Second Half) — Syntactic Analysis

1. What is structural ambiguity? Give an example.
2. What is shallow parsing (chunking)?
3. Define a Probabilistic Context-Free Grammar (PCFG).
4. What is a feature structure? Give an example.

Unit IV — Semantics and Pragmatics

5. List the requirements for a meaning representation.
6. What is First-Order Logic (FOL)? Give an example.
7. Represent the sentence “Every student reads a book” in First-Order Logic.
8. What is a word sense? Give an example.
9. Differentiate homonymy and polysemy with examples.
10. Define synonymy and hyponymy with examples.
11. What are thematic roles? Give an example.
12. What is Word Sense Disambiguation (WSD)?

Unit V — Discourse Analysis, Lexical Resources and Applications

13. What is discourse segmentation?
14. Differentiate anaphora and cataphora.
15. Differentiate stemming and lemmatization.
16. What is WordNet?
17. What is sentiment analysis? Give two real-world uses.
18. What is machine translation? Name its types.
19. What is a chatbot? Differentiate rule-based and AI-based chatbots.
20. List any four applications of Natural Language Processing.

PART B — 16 Mark Questions

Unit III (Second Half) — Syntactic Analysis

1. (a) Explain Probabilistic Context-Free Grammar (PCFG) with an example. (8)
   (b) Explain shallow parsing (chunking) with a chunk grammar and IOB tags. (8)
2. Explain feature structures and unification of feature structures with suitable examples.

Unit IV — Semantics and Pragmatics

3. Explain First-Order Logic as a meaning representation. Describe its elements (constants, functions, predicates, variables, connectives and quantifiers) with examples.
4. Explain the requirements for meaning representation and syntax-driven semantic analysis using semantic attachments.
5. Explain word senses and the relations between senses (homonymy, polysemy, synonymy, antonymy, hyponymy and meronymy) with examples.
6. Explain thematic roles and selectional restrictions with suitable examples.
7. Explain Word Sense Disambiguation using supervised, dictionary/thesaurus-based and bootstrapping methods.

Unit V — Discourse Analysis, Lexical Resources and Applications

8. Explain discourse segmentation, coherence and reference phenomena with suitable examples.
9. Explain the following NLP resources: Porter Stemmer, Lemmatizer, Penn Treebank, Brill's Tagger, WordNet, PropBank and FrameNet.
10. Explain sentiment analysis in detail — its levels, steps and approaches (lexicon-based, machine learning and deep learning using LSTM).
11. Explain machine translation and its approaches — rule-based, statistical and neural machine translation — with neat diagrams.
12. Explain chatbots — their types, architecture and working. Describe how a rule-based chatbot responds to user inputs.